



Secure Dev Labs

Ethical Hacking Submission File

Student Information

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Internship Track	Ethical Hacking	Week / Task	1 st
Submission Date	23/04/2026	Instructor	Muhammad Saad Rajput
Tools Used	Nmap	File Name	Muhammad Ahmed Raza_Lab Setup and Target Enumeration.pdf.

1. Task Overview

Task Title: Basic Network Scanning

Task Context: This phase of the project is centered on the architecture of a dedicated virtual laboratory intended for the systematic reconnaissance of a target infrastructure. The process encompasses the validation of network connectivity, the identification of active network nodes, and the detailed enumeration of accessible ports and running services. By doing so, we can effectively map the target's exposure and evaluate the complexity of its potential attack surface.

2. Objective

The core aim of this task is to deploy a fully operational testing environment, ensure seamless network communication, and execute comprehensive host discovery. Furthermore, the objective is to catalog all exposed services and open ports on the target machine to conduct a rigorous analysis of the security risks they present to the overall system integrity.

3. Lab Environment and Tools

Item	Details
Operating System	Kali Linux (virtual machine)
Target Machine	Metasploitable 2
Virtualization Tool	Virtual Box
Primary Tool	Nmap
Purpose	Lab setup, host discovery and service enumeration
Scan Command	<code>nmap -sV 192.168.56.102</code>
Target Type	Authorized internal lab target
Evidence	Screenshots are shown below

4. Steps Performed

- **Environment Setup:** Successfully deployed Kali Linux and Metasploitable 2 as virtual instances within VirtualBox, establishing a dedicated NAT network to facilitate secure inter-VM communication.
- **Connectivity Validation:** Confirmed stable network interaction between the attacker machine (192.168.1.107) and the target system (192.168.1.4) by executing ICMP echo requests via the ping utility.
- **Active Host Identification:** Conducted a network-wide discovery phase using the `nmap -sn 192.168.1.0/24` command to map out all currently reachable devices within the specified subnet.
- **Comprehensive Service Mapping:** Performed an intensive scan using `nmap -sS -p- -sV` on the target IP (192.168.1.4) to catalog every open port and determine the specific versions of the services currently exposed.

5. Evidence Section

Interns should insert clean, relevant screenshots in the same order as the steps above. Do not submit random images without captions.

Figure 1. Metasploitable 2 network setting

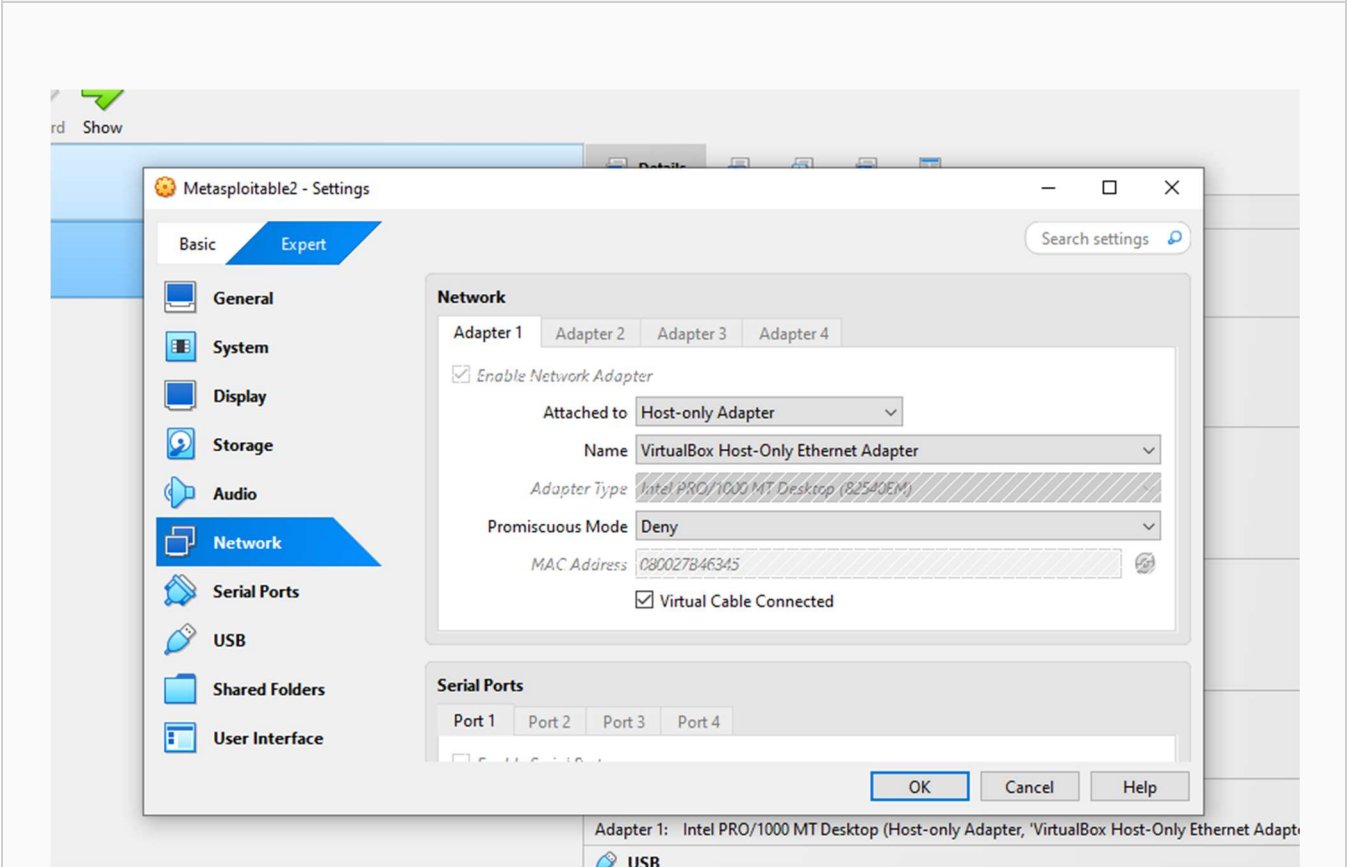


Figure 2. Kali Linux network interface setting

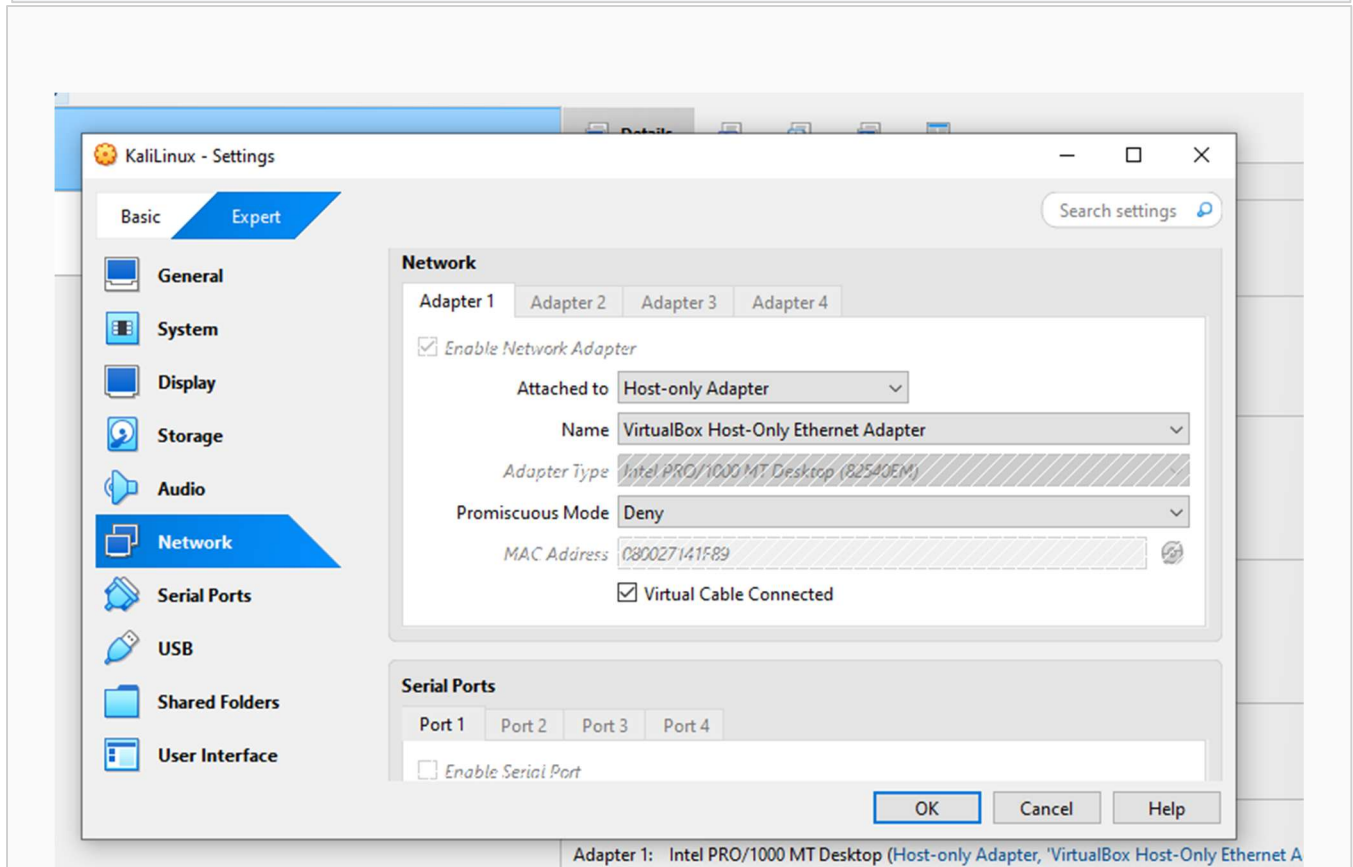
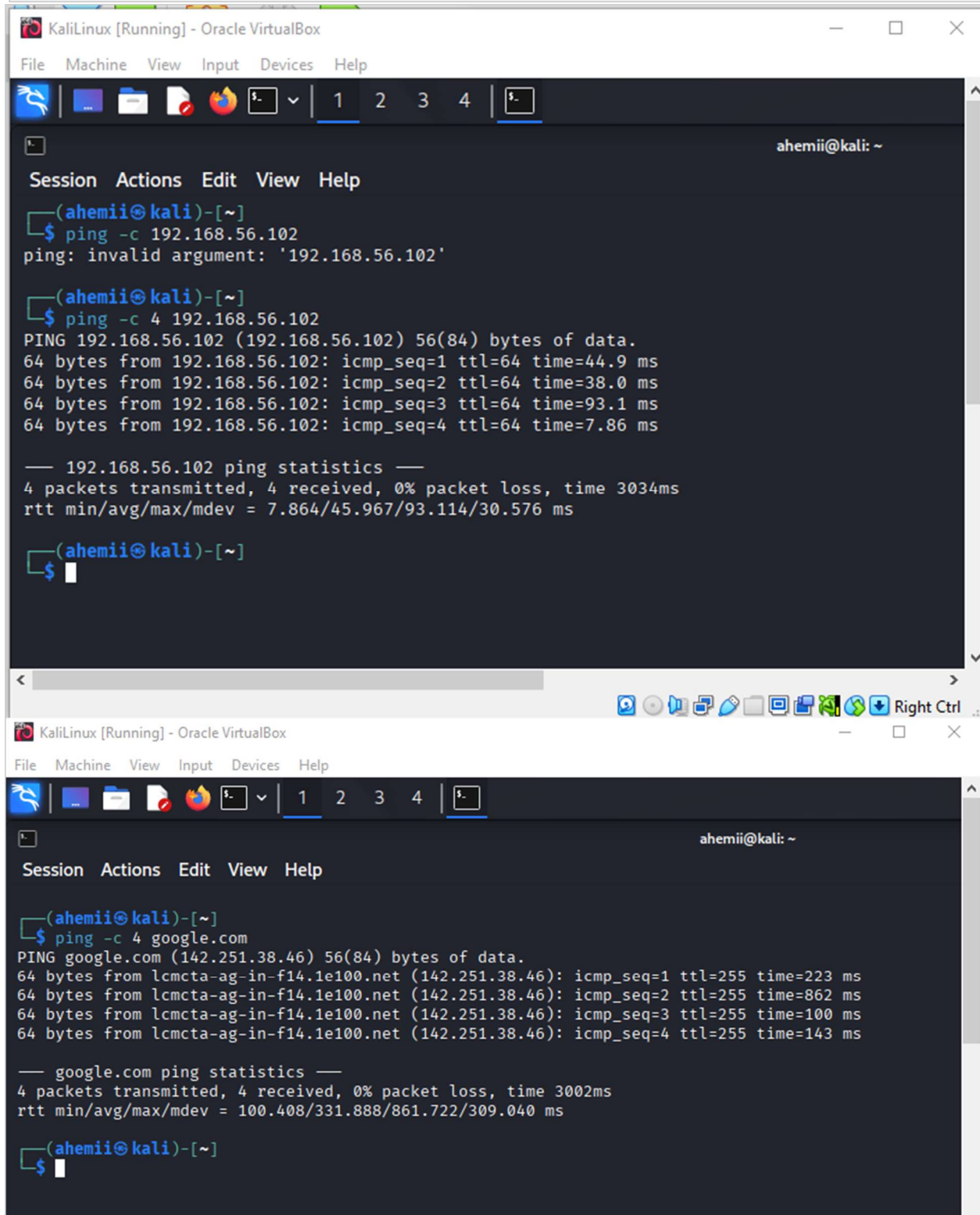


Figure 3. Kali Linux Internal and external connectivity



The image displays two screenshots of a Kali Linux terminal window running inside an Oracle VM VirtualBox. The window title is "KaliLinux [Running] - Oracle VirtualBox". The terminal shows the user "ahemii@kali" in the home directory. In the first screenshot, the user attempts to ping the IP address 192.168.56.102. The first command, `ping -c 192.168.56.102`, results in an error: "ping: invalid argument: '192.168.56.102'". The second command, `ping -c 4 192.168.56.102`, successfully executes, showing four successful ping requests with varying response times (44.9 ms, 38.0 ms, 93.1 ms, and 7.86 ms). The statistics show 4 packets transmitted, 4 received, 0% packet loss, and a total time of 3034ms. In the second screenshot, the user pings the domain "google.com" with the command `ping -c 4 google.com`. This also succeeds, showing four successful ping requests with response times of 223 ms, 862 ms, 100 ms, and 143 ms. The statistics show 4 packets transmitted, 4 received, 0% packet loss, and a total time of 3002ms.

```
KaliLinux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(ahemii@kali)~
$ ping -c 192.168.56.102
ping: invalid argument: '192.168.56.102'

(ahemii@kali)~
$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=44.9 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=38.0 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=93.1 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=7.86 ms

— 192.168.56.102 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3034ms
rtt min/avg/max/mdev = 7.864/45.967/93.114/30.576 ms

(ahemii@kali)~
$

KaliLinux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(ahemii@kali)~
$ ping -c 4 google.com
PING google.com (142.251.38.46) 56(84) bytes of data.
64 bytes from lcmcta-ag-in-f14.1e100.net (142.251.38.46): icmp_seq=1 ttl=255 time=223 ms
64 bytes from lcmcta-ag-in-f14.1e100.net (142.251.38.46): icmp_seq=2 ttl=255 time=862 ms
64 bytes from lcmcta-ag-in-f14.1e100.net (142.251.38.46): icmp_seq=3 ttl=255 time=100 ms
64 bytes from lcmcta-ag-in-f14.1e100.net (142.251.38.46): icmp_seq=4 ttl=255 time=143 ms

— google.com ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3002ms
rtt min/avg/max/mdev = 100.408/331.888/861.722/309.040 ms

(ahemii@kali)~
$
```

Figure 4. Live Host Enumeration

```
(ahemii@kali)-[~]
$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=98.1 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=27.5 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=64.0 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=7.13 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=70.9 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=103 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=11.0 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=40.5 ms
64 bytes from 192.168.56.102: icmp_seq=9 ttl=64 time=72.9 ms
64 bytes from 192.168.56.102: icmp_seq=10 ttl=64 time=20.7 ms
64 bytes from 192.168.56.102: icmp_seq=11 ttl=64 time=10.5 ms
^C
— 192.168.56.102 ping statistics —
12 packets transmitted, 11 received, 8.33333% packet loss, time 11058ms
rtt min/avg/max/mdev = 7.125/47.825/102.705/33.799 ms
```

Figure 5. Active ports and Services Enumeration

```
KaliLinux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(ahemii@kali)-[~]
$ nmap -sV 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-18 17:00 PKT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
Nmap scan report for 192.168.56.102
Host is up (0.79s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp?
22/tcp    open  ssh?
23/tcp    open  telnet?
25/tcp    open  smtp?
53/tcp    open  domain?
80/tcp    open  http          Apache httpd 2.2.8 (DAV/2)
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn?
445/tcp   open  microsoft-ds?
512/tcp   open  exec?
513/tcp   open  login?
514/tcp   open  shell?
1099/tcp  open  rmiregistry?
1524/tcp  open  ingreslock?
2049/tcp  open  nfs           2-4 (RPC #100003)
2121/tcp  open  ccproxy-ftp?
3306/tcp  open  mysql?
5432/tcp  open  postgresql?
5900/tcp  open  vnc?
6000/tcp  open  X11?
6667/tcp  open  irc           UnrealIRCd
8009/tcp  open  ajp13?
8180/tcp  open  unknown
MAC Address: 08:00:27:B4:63:45 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Host: metasploitable.localdomain

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 283.75 seconds
```


6. Findings and Analysis

Port	Service	Observation	Possible Risk
21/tcp	FTP (vsftpd 2.3.4)	FTP service is running and accessible	Known backdoor vulnerability in vsftpd 2.3.4 may allow remote access
22/tcp	SSH	Remote login service detected	Weak credentials may allow brute-force attacks
23/tcp	Telnet	Unencrypted remote access service	Credentials can be intercepted easily (plaintext transmission)
25/tcp	SMTP	Mail server is active	Can be used for spam relay or user enumeration
53/tcp	DNS (BIND)	DNS service is exposed	May allow DNS spoofing or zone transfer attacks
80/tcp	HTTP (Apache)	Web server is running	Possible web vulnerabilities (XSS, SQLi, misconfigurations)
111/tcp	RPCBind	RPC service is active	Can expose network services and aid in enumeration
139/tcp	NetBIOS	File sharing service detected	May allow unauthorized access to shared resources
445/tcp	SMB (Samba)	SMB service is running	Vulnerable to SMB exploits (e.g., EternalBlue)
512/tcp	Exec (rexecd)	Remote execution service	Can allow unauthorized command execution

513/tcp	Login (rlogin)	Remote login service	Weak authentication can lead to unauthorized access
514/tcp	tcpwrapped	Service is filtered/protected	May still indicate hidden or restricted services
1099/tcp	Java RMI	Java remote method invocation service	Can be exploited for remote code execution
1524/tcp	Bind Shell	Root shell service detected	Direct root access without authentication (critical vulnerability)
2049/tcp	NFS	Network file system service	Unauthorized file access or data leakage
2121/tcp	FTP (ProFTPD)	Secondary FTP service detected	May contain misconfigurations or weak authentication
3306/tcp	MySQL	Database service is running	Weak credentials may expose sensitive data
3632/tcp	Distccd	Distributed compiler service	Known RCE vulnerability in older versions
5432/tcp	PostgreSQL	Database service detected	Unauthorized access may expose database contents
5900/tcp	VNC	Remote desktop service	Weak/no authentication can allow full system access
6000/tcp	X11	Display server detected	Can allow remote GUI access if misconfigured
6667/tcp	IRC	IRC service is running	May be used for botnet communication

6697/tcp	IRC (SSL)	Secure IRC service detected	Still may allow unauthorized communication channels
8009/tcp	AJP13	Apache JServ protocol	Vulnerable to Ghostcat attack (file inclusion)
8180/tcp	HTTP (Tomcat)	Application server detected	Default credentials or misconfigurations possible
8787/tcp	DRb (Ruby)	Ruby distributed object service	May allow remote code execution
39633/tcp	Mountd	NFS mount service	Can expose file system access
43192/tcp	Java RMI	Secondary RMI service	Potential RCE attack vector
43259/tcp	Nlockmgr	Network lock manager	Can be used in NFS-related attacks
53294/tcp	Status	RPC status service	Provides system information useful for attackers

7. Short Conclusion

Analysis of the scan data indicates a critical lack of security hardening on the target system. By exposing a wide array of services like SMB, RPC, and vsftpd, the system's attack surface has been significantly widened. The discovery of a bind shell on port 1524 is a definitive 'red flag,' indicating a likely prior compromise or a severe configuration flaw. These findings suggest that the target lacks basic defense-in-depth measures, making it an easy target for remote exploitation. We can conclude that the system's current security posture is inadequate against even basic automated attack vectors.